

and I'll present them as examples here.

I have written two full-length books using word-processors and have struggled with the problems of managing a huge document—for example, changing the formatting or layout uniformly is a big problem. Then I discovered LaTeX, which is useful for creating professional-quality (print quality) documents like books. It is not a tool but a language (a markup language like HTML) where I can write the text with markups, and compile the code and run it to get a PDF or DVI output. And yes, we will get compiler warnings and errors, and will need to hunt for and fix them. Is it worth the effort to write code for something as simple as document formatting? In my experience, it's well worth it and it even saves considerable time! For example, with LaTeX, creating and updating an index for a book is straightforward and the results are of high quality. I have spent days struggling to do that in word processors.

I recently faced the problem of creating diagrams. How do you generate graphs and diagrams for your book, article or that very-important presentation to your CEO? It is fairly straightforward to use a drawing application to create diagrams or graphs. My hand-drawn diagrams look very bad. To create graphs, yes, we can put the data in a spreadsheet and create the output as a graph. But is it of high quality? No! What if you want the data imported and updated automatically as your data changes, and plotted as a graph? 'Automatic' often means it is done programmatically—and that means using a language! You might shudder—writing programs to create diagrams? Yes, for example, GraphViz (short for Graph Visualisation) is a popular set of open source tools for drawing graphs using the DOT language. I am yet to try using it for complex graphs, but I have had a good experience using LaTeX, so I think I'll make best use of it.

DSLs are most effective for solving problems in specific domains. Imagine creating graphs by writing programs in C++—it doesn't make much sense and it's better to use GraphViz. But the problem is that DSLs will never be mainstream languages, while general-purpose languages, by the virtue of their generality, will be the most popular languages.

Which is the 'programming language'?

Most programmers know which the best programming language is. The only problem is that the best programming language differs for each programmer! "It is C you know—there is nothing like that"; "It's definitely Python—it's so intuitive to program"; "It's PHP—I think you haven't used it. If you do, you'll agree that it's the best"...

In other words, programmers think that their favourite language is the best. But then, there are so many *bests*...

What does 'the language of the year' mean?

While there is no 'best' language, there can be one that is very popular, is very widely used, or that has received the

most attention. We can use these factors as the basis to selecting a language for our award.

Tiobe [www.tiobe.com] Programming Community Index (TPCI) is widely referred to in relation to the popularity of programming languages. It is updated every month. It calculates the ratings for a language based on the number of hits that a language gets in popular search engines, including Google, Yahoo! and Wikipedia.

The following table rates top 10 languages for 2009 (updated for December).

Language	Ratings
Java (-2.31%)	17.061%
C (+0.12%)	16.285%
PHP (+0.29%)	9.770%
C++ (-1.72%)	9.175%
(Visual) Basic (-1.70%)	7.778%
C# (+1.61%)	6.258%
Python (+0.62%)	5.185%
JavaScript (+0.45%)	3.515%
Perl (-0.91%)	2.692%
Ruby (+0.34%)	2.653%

In this table, the language and its popularity ratings in percentage are shown. In brackets is the difference in popularity from last year. For example, Java's popularity was more last year (19.3 per cent) and it has dropped to 17.061 per cent.

What strikes you is C's popularity. C, which is more than three decades old, holds the second spot, and it is almost as popular as Java. C is still the most popular language for embedded systems (though Java is catching up fast in this area). With mobiles and embedded devices becoming more widely used than ever before, C is gaining popularity.

C++ has steadily lost its popularity—its rating was 15.5 per cent in 2001, and now it's just 9.175 per cent. Its popularity might rise again when the next standard (C++0x) gets released. Languages like C# and Java have eaten into C++'s pie.

According to Tiobe, this year, 'company-related' languages C# (Microsoft), ActionScript (Adobe), and Objective-C (Apple) have gained popularity—all three are in the top 20.

Object-oriented languages are the most popular (55.1 per cent), followed by procedural languages (39.9 per cent). As a whole, these two categories are the most popular, leaving a very small share (5 per cent) to functional and logical paradigms.

Statically-typed languages are most popular with a